Earth and Planetary Science Letters 272 (2008) 464–475

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

journal homepage: www.elsevier.com/locate/epsl

Reassembling the Paleogene–Eocene North Atlantic igneous province: New paleomagnetic constraints from the Isle of Mull, Scotland

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ARTICLE INFO

Article history:
Received 20 September 2007
Received in revised form 5 May 2008
Accepted 9 May 2008
Available online 28 May 2008

Editor: R.W. Carlson

Keywords:
paleomagnetic poles
North Atlantic Igneous Province
British Tertiary Igneous Province
tertiary

ABSTRACT

The paleomagnetic data sets from the British Tertiary Igneous Province (BTIP) have recently been criticized as being unreliable and discordant with data from elsewhere in the North Atlantic Igneous Province (NAIP) [Riisager et al. Earth Planet. Sci. Lett. 201 (2002) 261–276; Riisager et al. Earth Planet. Sci. Lett. 214 (2003) 409–425]. We offer new paleomagnetic data for the extensive lava flow sequence on the Isle of Mull, Scotland, and can confirm the paleomagnetic pole positions emanating from important earlier studies. Our new north paleomagnetic pole position for Eurasia at 59±0.2 Ma has latitude 73.3°N, longitude 166.2°E (dp/dm=5.2/7.0).

A re-evaluation and an inter-comparison of the paleomagnetic database emanating from the NAIP were carried out to test for sub-province consistency. We find a general agreement between the Eurasian part of NAIP (BTIP and Faeroes) and East Greenland data. However a compilation of West Greenland data displays a large and unexplained dispersion. We speculate on if this is related to different sense of block rotation of the Tertiary West Greenland constituents. Combining all data from the NAIP constituents, give a pole position at 75.0°N, 169.9°E (N=25, K=84.3, A_{95}=3.2) in Eurasian reference frame.

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1. Introduction

The North Atlantic Igneous Province (NAIP), now straddling the North Atlantic Ocean and the West Greenland–Baffin corridor (Fig. 1) is the result of widespread Tertiary volcanism and is one of the well-known Large Igneous Provinces (LIP; Courtillot et al., 1999, Saunders et al., 1997). The NAIP was split in two by the opening North Atlantic Ocean, and the principal components of the province are now exposed in East and West Greenland, on the Faeroe Islands and the British Isles. The bulk of the British part of the NAIP, known as the British Tertiary Igneous Province (BTIP), crops out in Antrim (Northern Ireland) and on the Isles of Mull and Skye in Scotland (Fig. 1; Saunders et al., 1997). Most of the syn-breakup volcanic material imaged as seaward dipping reflectors within the Eurasian and Greenland margins is also presumed to belong to the NAIP (Saunders et al., 1997).

With a volume estimate between 1.8×10^6 km^3 (Eldholm and Grue, 1994) and 3.7×10^6 km^3 (Holbrook et al., 2001), the NAIP is one of the most voluminous of the post-Late Jurassic (150 Ma) LIPs. Its origin has been linked to the Icelandic plume through its close spatial relationship (Lawver and Müller, 1994, Torsvik et al., 2001) and through its geochemical characteristics (Fitton et al., 1997, Graham et al., 1998, Kempton et al., 2000, Kerr, 1995) with additional supporting arguments in Campbell (2007). However, alternative mechanisms for the formation of LIP's such as edge-driven flow (King and Anderson, 1998) have also been forwarded and the origin of the NAIP remains an ongoing subject of lively debate (Foulger, 2005, Meyer et al., 2007).

The age and paleomagnetic signature of the various parts of the NAIP have been investigated since the 1950's. The ages of the pre-Atlantic breakup volcanics are converging on the range of 62–58 Ma (summary in Hansen et al., 2002, Sinton et al., 1998), while a second and more voluminous phase of magmatism, dated to 56–53 Ma coincides with the opening of the north Atlantic (Saunders et al., 1997) at the time of magneto-chron C24r (~55 Ma, Cande and Kent, 1995). Products of this pulse are exposed in East Greenland (Larsen and Saunders, 1998, Tegner et al., 1998) and the Faeroe Islands (Larsen et al., 1999). Recent magneto-stratigraphic studies on the NAIP were carried out on the extensive lava sequences in Western Greenland (Riisager et al., 2003a,b, Riisager and Abrahamsen, 1999) and the Faeroe Islands (Riisager et al., 2002). The lava sequences of the BTIP on the other hand, have not received much attention paleomagnetically since the 1970's (Hall et al., 1977, Løvlie et al., 1972, Mussett et al., 1980, Wilson, 1970).

Following their work in Western Greenland (Riisager et al., 2003a,b) and the Faeroes Islands (Riisager et al., 2002), Riisager et al. (2003b) noted similarity between the ages and paleomagnetic poles from the two sub-provinces after correcting for the younger opening of the

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doi:10.1016/j.epsl.2008.05.016

North Atlantic. However, Riisager et al. (2002) observed that the British paleomagnetic data differed significantly from their Faeroes result, even though both reside in on the same plate. Given that many of the paleomagnetic studies in the BTIP were carried out 20 to 30 years ago using less precise laboratory instrumentation and sometimes less thorough procedures for differentiating between magnetizations of different ages, they attributed the discordance to the lower reliability of the British data. We found this conclusion surprising and set out to re-examine the paleomagnetic record carried by the principal lava flow sequences of the BTIP. Doing so, we provide new paleomagnetic poles to merit a fresh intercomparison between NAIP poles. All three of the principal lava sequences of the BTIP have been re-sampled for paleomagnetic analysis and radiometric dating. Here we report new results from the Isle of Mull lava (Fig. 1). The studies on the Isle of Skye and in Antrim will be reported separately. With the exception of the study of Mussett et al. (1980) that focused on the intrusive centres of Mull, the present investigation is the first paleomagnetic study of the lava sequences on Mull since Hall et al. (1977).

2. The Mull Lava Group

The Isle of Mull preserves one of three major sequences of Tertiary lavas in the British Tertiary Igneous Province (Fig. 1). The lava sequence is collectively termed the Mull Lava Group (MLG) and rests on a laterally persistent mudstone horizon known as the Gribun Mudstone Member (e.g. Emeleus and Bell, 2005). The Group is made up of tholeiitic basalts of the Staffa Lava Formation, overlain by predominately olivine-basalts of the Mull Plateau Lava Formation (MPLF; Emeleus and Bell, 2005). The age of MPLF is constrained by several radiometric data, the most recent being from Chambers and Pringle (2001), using the $^{40}Ar/^{39}Ar$ method. These authors obtained an age of 60.6±0.3 Ma (all errors reported as 1σ throughout the paper) for a lava flow near the base of the Formation. This age is consistent with those of Mussett (1986). Chambers and Pringle (2001) also report a weighted mean age of 58.4±0.2 Ma for two lava samples taken at an elevation of 700 m above sea level (Fig. 1), but argue that this age resemble a hydrothermal event and should be used with caution. However, this age is supported by our preliminary $^{40}Ar/^{39}Ar$ age of 58.7±1.1 Ma (inverse isochron) for a lava flow in the same sequence.

The entire MLG is cut by three major intrusive centres comprising gabbros, granophyres, ring dykes and granites (Emeleus and Bell, 2005). The intrusive centres are offset and aligned from oldest to youngest along a northwesterly axis, parallel to the dominant dyke trend (Fig. 1). Some age constraints are given by Chambers and Pringle (2001) on the intrusive units and indicates a cooling age of 58.5±0.2 Ma for the latest magmatic event in the Centre 3 (Lock Ba ring dyke). However, Mussett (1986) obtained an age of 56.5±1.0 for the same ring dyke.

Emplacement of the intrusive complexes was followed by extrusion of olivine-poor tholeiitic lavas known as the Mull Central Lavas (MCL). These pillowed flows are presumed to have been emplaced in water filled calderas associated with the intrusive centres (Bailey et al., 1924) and a best age estimate for their extrusion is given by a inverse isochron of 57.9±1.2 Ma (Chambers and Pringle, 2001).

Zeolites are widespread in MLG and their composition is thought to be related to depth of burial. The presence of laumonite indicates that the MPLF had a thickness over 2000 m (Walker, 1971) of which 1000 m is preserved. A hydrothermal alteration zone (epidote and prehnite; Fig. 1) is superimposed on these zeolites zones and is related to circulating water and heat from the intrusive Centres (Walker, 1971).

NW–SE to N–S trending dykes and arrays of plugs are characteristic of the BTIP (Fig. 1; Speight et al., 1982). These are believed to have functioned as feeders for the lava fields (Kerr, 1997). The majority of the dykes in Mull post-date the lavas and are in greatest abundance close to the central complexes and are contemporaneous with their emplacement (Jolly and Sanderson, 1995). Chambers and Pringle (2001) get a weighted mean age of 58.2±0.1 for dykes cutting the Lock Ba ring dyke indicating that the dykes are temporally associated to the central complexes. Many dykes do post-date the central complexes and are unaffected by the hydrothermal alteration zone (e.g. Emeleus and Bell, 2005, p.81).

The lava fields on Mull form an open syncline, with an axis parallel to the main dyke orientation (NW–SE; Bailey et al., 1924). Dips of flows are generally less than 5°. Locally, lava flows are tilted independently within fault-bounded blocks.

3. Paleomagnetic sampling

The Mull Plateau Lava Formation (MPLF) constitutes the larger part of the Mull Lava Group and earlier paleomagnetic investigation of this Formation uncovered primary remanent magnetisation (Ade-Hall et al., 1972, Hall et al., 1977). We focused our sampling on this same sequence of lavas. A total of 254 25 mm diameter drill cores were extracted with a gasoline portable drill at 28 sites in the lava sequence during the summer of 2004 (MPLF in Table 1; Fig. 1). Three of the sampling sites encompass two sampled flows (M5, M10 and M28) making the total number of flows sampled 31. Cores were extracted from grey lava flow interiors and, when exposed, from reddened lava flow tops. Special effort was made to achieve a horizontal and vertical spread in sampling sites to examine the effects of tilting across the island and to obtain stratigraphic control on the paleomagnetic record. Four paleomagnetic contact tests (at sites M1, M2, M3 and M4) were performed where dykes cut the lava sequence. A vertical sampling profile was conducted southwest of Lock Na Keal starting at site M5 (103 m above sea level) where the lava sequence rests on the Gribun Leaf Bed, and ending at site M31 some 311 m above sea level.

Cursory sampling was carried out at three additional sites (M13, M14 and M15) within the intrusive complexes (Table 1, Fig. 1). M14 includes a contact test based on a dyke cutting a gabbroic body.

4. Paleomagnetic analysis

Paleomagnetic laboratory experiments were carried out at the Geological Survey of Norway (NGU) and the University of Bergen (Norway). Natural remanent magnetization (NRM) was measured using AGICO JR6A and JR5A spinner magnetometers, mounted within Helmholtz coil systems. Components of magnetisation were identified using stepwise thermal and alternating field demagnetisation techniques. Thermal demagnetisation proved most successful and was mainly used. In all, 319 specimens were thermally demagnetised using around 15–25 heating steps, and 34 specimens, at least one from each sampling site, were ubjected to alternating field demagnetisation up to 95 mT. The directions and unblocking temperature spectra of characteristic remanence components (ChRM) were determined using the LineFind algorithm of Kent et al. (1983) as implemented in the Super-IAPD program (available at www.geodynamics.no; Torsvik et al. 2000).

4.1. Mull plateau lava formation

NRM intensities for the MPLF range between 20 and 15000 A/m but are commonly around 7000 A/m. The majority of specimens carry a single vector component of NRM and typical thermal demagnetisation behaviour for the lava is illustrated in Fig. 2a. The rate of remanence unblocking increases towards heating steps near 530 °C, then again around the Curie point of magnetite (578 °C) and around 640 °C. The magnetic remanence is carried by Fe-rich titanomagnetite and in 20% of the samples, additional titanohematite.

Demagnetisation results for a typical dyke specimen are illustrated in Fig. 2b. Again, mineral phases with different thermal stabilities carry a single vector component of remanence. The unblocking temperature spectrum in Fig. 2b suggests a variety of magnetite/titano-magnetite carriers — no hematite.

All the characteristic component directions obtained from the lava are of reverse polarity (Table 1; Fig. 3a). This was also observed by Hall

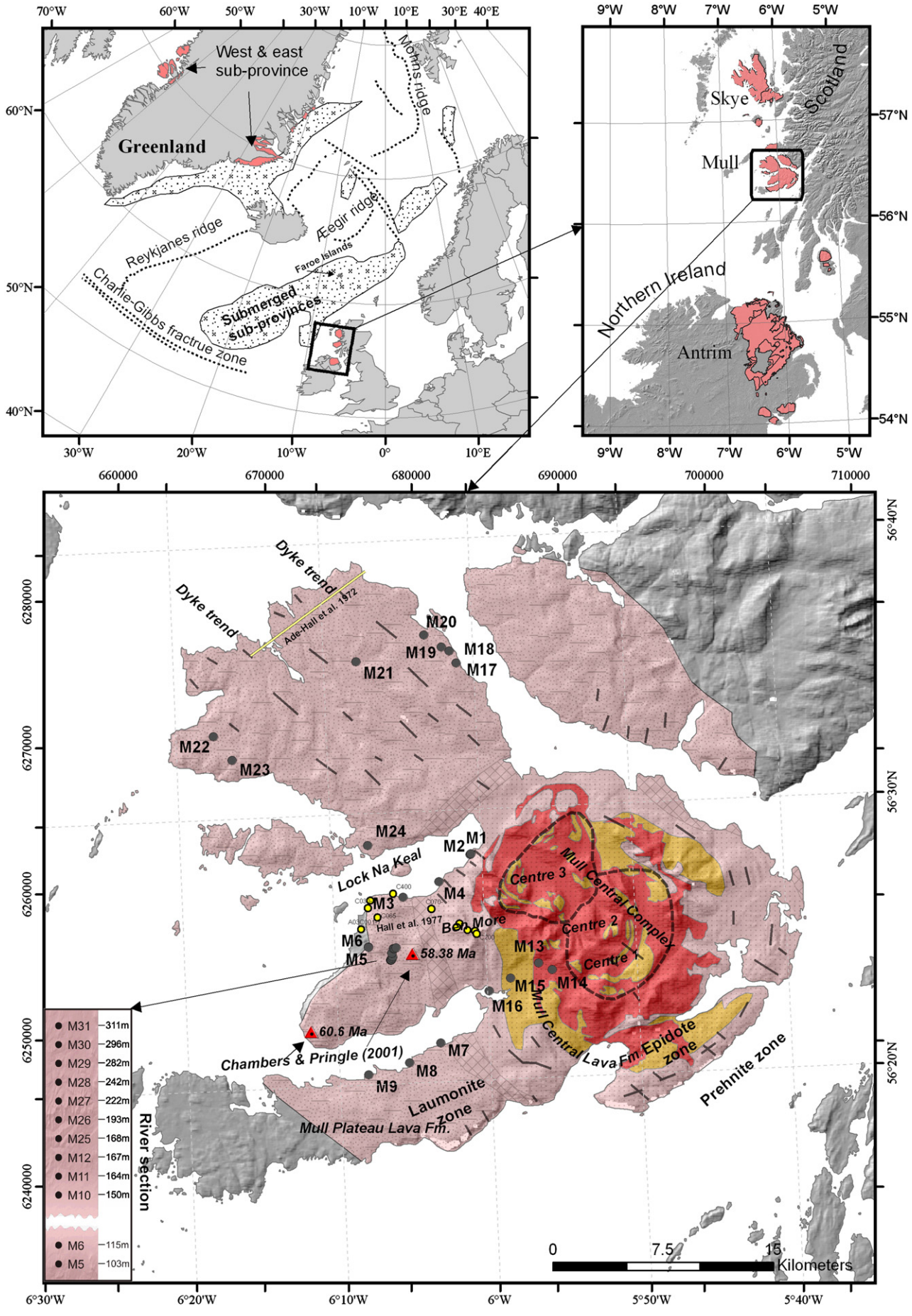

West & east sub-province
Greenland
Mohns ridge
Ægir ridge
Reykjanes ridge
Charlie-Gibbs fractrue zone
Faroe Islands
Submerged sub-provinces
Skye
Scotland
Mull
Northern Ireland
Antrim
Dyke trend
Ade-Hall et al. 1972
M20
M19
M18
M17
M21
M22
M23
M24
M2
M1
Lock Na Keal
M4
M3
Hall et al. 1977
Ben More
M6
M5
58.38 Ma
60.6 Ma
Chambers & Pringle (2001)
M7
M8
M9
Laumonite zone
Mull Plateau Lava Fm.
Centre 3
Centre 2
Centre 1
Mull Central Complex
M13
M15
M14
M16
Mull Central Lava Fm
Epidote zone
Prehnite zone
River section
M31 311m
M30 296m
M29 282m
M28 242m
M27 222m
M26 193m
M25 168m
M12 167m
M11 164m
M10 150m
M6 115m
M5 103m
0 7.5 15 Kilometers

Table 1
Paleomagnetic sampling sites and results

Site	Lithology	Comment	X (m)	Y (m)	D	I	D_{str}	I_{str}	N/n	R	k	α_{95}	Polarity
M1	MPLF	EZ	684077	6262810	324.6	56.8	324.6	56.8	11/12	10.96	262.79	2.8	N
	Dyke	CT, EZ	–	–	330.9	57.7	330.9	57.7	8/8	7.98	303.95	3.2	N
M2	MPLF	EZ	684012	6262744	307.9	68.1	307.9	68.1	5/12	4.98	262.78	4.7	N→R
M3	Dyke	CT, EZ	–	–	323.9	70.6	323.9	70.6	6/6	6.00	1038.85	2.0	N
	MPLF	EZ	679428	6259824	147.1	−55.8	147.1	−55.8	4/6	3.99	258.91	5.7	R
	Dyke	CT, EZ	–	–	154.6	−29.4	154.6	−29.4	5/6	4.99	321.21	4.3	R
M4	MPLF	EZ	681887	6260891	175	−32.4	175	−32.4	8/10	6.98	273.80	2.4	R
	Dyke	CT, EZ	–	–	162.3	−46.4	162.3	−46.4	7/12	7.99	514.18	3.7	R
M5*	MPLF A*	Base of section (103m)	677081	6256431	155.9	−54.2	155.9	−54.2	4/5	3.99	242.19	5.9	R
	MPLF B*	~103m	–	–	181.8	−46.1	181.8	−46.1	5/6	4.99	424.98	3.7	R
M6*	MPLF	Above M5 (115m)	677084	6256406	161.8	−59.1	161.8	−59.1	5/6	5.00	2537.85	1.5	R
M7*	MPLF	USP	682008	6249869	170.1	−60.8	190.3	−61.9	7/9	6.99	437.90	2.9	R
M8*	MPLF	USP	679879	6248511	190.9	−62.3	204.2	−62.0	8/8	7.98	336.92	3.0	R
M9*	MPLF	USP	677109	6247675	175.3	−50.0	182.5	−51.1	9/10	8.96	218.00	3.5	R
M10*	MPLF A*	Above M6 (150m)	678585	6255514	164.7	−42.4	166.1	−46.2	3/3	4.00	1252.3	2.6	R
	MPLF B*		–	–	171.1	−56.8	174.1	−60.3	3/3	2.00	1172.8	7.3	R
M11*	MPLF	Above M10 (164m)	678594	6255517	165.9	−43.9	167.4	−47.4	7/9	6.96	150.24	4.9	R
M12*	MPLF	Above M11 (167m)	678625	6255547	178.3	−48.7	181.0	−52.0	4/6	3.99	249.11	5.8	R
M13	Granophyre	Center 1, EZ	688704	6255338	208.6	−33.6	208.6	−33.6	6/6	9.94	145.87	4.0	R
M14	Gabbro	Center 1, EZ	689631	6254872	162.1	−48.0	162.1	−48.0	7/8	6.95	163.68	4.3	N→R
	Dyke	CT, EZ	–	–	158.6	−62.8	158.6	−62.8	7/16	7.96	135.42	4.4	R
M15	Granophyre	Center 1, EZ	686780	6254301	153.1	−63.8	184.4	−64.1	9/9	15.85	100.62	3.7	R
M16*	MPLF	USP, EZ	685317	6253433	155.7	−57.0	179.4	−57.7	7/8	6.97	191.88	4.4	R
M17*	MPLF	USP, EZ	683060	6275844	191.8	−47.5	193.1	−50.3	7/10	6.95	125.51	5.4	R
M18*	MPLF	USP	682585	6276670	185.2	−62.7	197.1	−68.7	8/9	7.97	213.69	3.8	R
M19*	MPLF	USP	682048	6276920	176.4	−63.2	184.0	−66.7	9/10	15.93	216.95	2.5	R
M20*	MPLF	USP	680852	6277747	175.6	−72.3	188.4	−75.7	10/10	9.96	251.35	3.1	R
M21*	MPLF	USP	676223	6275920	187.3	−67.8	187.3	−67.8	9/10	10.99	786.16	1.6	R
M22*	MPLF	USP	666493	6270820	201.6	−61.0	203.2	−53.1	8/11	7.96	193.88	4.0	R
M23*	MPLF	USP	667766	6269194	200.2	−71.9	203.1	−64.0	5/8	4.99	308.17	4.4	R
M24	MPLF	USP Lightning strike	677020	6263365	–	–	–	–	0/8				
M25*	MPLF	Above M12 (168m)	678651	6255560	167.6	−50.2	169.7	−53.9	7/8	8.94	135.42	4.4	R
M26*	MPLF	Above M25 (193m)	678683	6255642	156.8	−56.1	158.2	−60.0	7/7	7.98	444.49	2.6	R
M27*	MPLF	Above M26 (222m)	678723	6255758	190.2	−65.1	197.1	−67.7	7/8	7.98	293.89	3.2	R
M28*	MPLF A*	Above M27 (242m)	678657	6256012	227.8	−64.9	236.1	−65.1	5/5	5.00	1249.00	2.2	R
	MPLF B*	Above M27 (242m)	–	–	198.7	−70.8	209.2	−72.9	6/6	6.00	1766.00	1.6	R
M29*	MPLF	Above M28 (282m)	678785	6256326	160.4	−42.2	161.5	−46.0	8/9	7.99	1008.42	1.7	R
M30*	MPLF	Above M29 (296m)	678865	6256329	173.9	−37.0	175.5	−40.5	6/8	10.63	26.85	9.0	R
M31*	MPLF	Above M30 (311m)	678951	6256384	202.4	−47.2	206.3	−49.2	8/8	11.96	245.35	2.8	R
Mean remanence direction for MPLF in situ			–	–	177.5	−57.3	–	–	26	25.3	35.9	4.8	R
Mean remanence direction for MPLF corrected			–	–			−182.9	−58.9	26	25.3	36.9	4.7	R

X/Y = UTM zone 29 coordinates of sampling sites (datum WGS84); D/I = declination/inclination of flow mean remanence directions; D_{str}/I_{str} = tilt corrected directions; N/n = number of remanence directions/specimens; R = length of resultant vector; k and α_{95} are the Fisher (1953) precision parameter and half angle of the cone of 95% confidence. N→R denotes change from normal to reverse polarity at higher unblocking temperatures. Lava flow results marked with an asterisk (*) are use in calculation of the overall mean direction at the bottom of the table. Abbreviations in the "Comment" are: EZ = epidote zone; CT = contact test, USP = uncertain stratigraphic position; Site elevations above the datum (WGS1984) are given in parentheses; "Centre 1" refers to the volcanic centres shown in Fig. 1.

et al. (1977) who sampled a longer traverse though the same lava sequence. Samples at Site M24 gave anomalous results and contained very high NRM intensities. Combining results from thermal and AF demagnetisation we interpret this site as affected by a lightning strike.

The age of the remanence carried by lava flows with respect to dykes cutting the flows can be established from the four paleomagnetic contact tests carried out at M1, M2, M3 and M4. Sites M1 and M2 lie close together within the epidote alteration zone around the intrusive complexes (Fig. 1; Walker, 1971). The dykes at both sites carry normal polarity remanence (Figs. 2b and 3b). The intruded lava flow at M1 preserves only normal polarity remanence, as does much of the enclosing lava at M2. Significantly, a single specimen 76 cm away from the dyke at M2 carries a reversed remanence with high unblocking temperatures (Fig. 2c). It would appear that the lava flows at M1 and M2 are largely remagnetised in a normal polarity field, due to the close proximity of numerous small dykes, the possible presence of unrecognised or hidden intrusives, or due to their positions within the alteration zone close to the younger intrusive complexes. Nevertheless, the detection of the reversed polarity remanence at high unblocking temperatures at M2 hints at a pre-dyke age for the reversed polarity remanence.

The dykes at M3 and M4 carry exclusively reversed polarity remanence and so do the enclosing lava flows. Nevertheless, the dyke and lava directions are oblique and the dyke directions gives way to the lava direction close to a half dyke width into the heated lava flows. The contact tests at M3 and M4 are, therefore, both positive and indicate pre-dyke ages for the remanence carried by the lavas at the two localities.

4.2. Intrusive complexes

Samples from sites M13, M14 and M15 were drilled in the intrusive complexes (Fig. 1; Table 1). The granophyre at site M13, probably part

Fig. 1. Outline of the North Atlantic Igneous Province, British Tertiary Igneous Province and the geology of Mull. Submerged sub-provinces are taken from Larsen and Saunders (1998). The Tertiary rocks of Mull are divided into the Mull Plateau Lava Fm. (pink), Mull Central Lavas (yellow) and the Mull Central Complex (red). Laumonite, prehnite and epidote zones are the metasomatic zones defined by Walker (1971). Map of Mull: Universal Transverse Mercator projection, zone 29, distance units in metres. The sampling traverse of Ade-Hall et al. (1972) is shown by the yellow line and the sampling localities of Hall et al. (1977) are shown by yellow dots.

of Centre 1 (British Geological Survey, 1992), carries a high-temperature reverse polarity remanence direction and a normal polarity overprint stable up to 350 °C. Site M14 is a gabbroic body of Centre 1, cut by a 1.4 m wide dyke. The magnetic signature of M14 gabbro samples show a complex pattern related to several thermoremanent magnetisation (TRM) events in the overall signature, however the original TRM of the gabbro is of a reverse direction, unblocked in the range of 570–590 °C (Fig. 2d (1)). A normal overprint is unblocked at 10–570 °C (Fig. 2d (2)), which could be related to regional heating associated with the emplacement to intrusion of Centre 2. Dagley et al. (1987) showed some cone sheets of Centre 2 to be of normal polarity. The dyke carries a reverse polarity remanence (Fig. 3b) and induces a lower temperature (150–490 °C) reverse polarity TRM direction in the gabbro close to the dyke (Fig. 2d (3)). For the gabbro, a normal direction is also seen in the highest temperature range but is interpreted to a chemical remanent magnetisation (CRM; Fig. 2d (4)). Site M15 is located in the MCL (Fig. 1; British Geological Survey, 1992) but actually consists of a reversely magnetized granophyre (Table 1). A normal component overprint is seen in the lower temperature range (<300 °C).

5. Paleomagnetic pole

Results from samples in the intrusive complexes at site M13, M14 and M15 are few in number and restricted in scope and are not considered further. Contact tests at M2, M3 and M4 imply a pre-dyke age for the remanence in the lavas and, like Hall et al. (1977) and Ade-Hall et al. (1971) we assign a primary or near primary origin to this remanence. In all, 26 of the sampled lava flows in the MPLF carry a reliable record of the paleofield at the time of their emplacement (Fig. 3a; Table 1). The flow-mean directions are well grouped and yield an

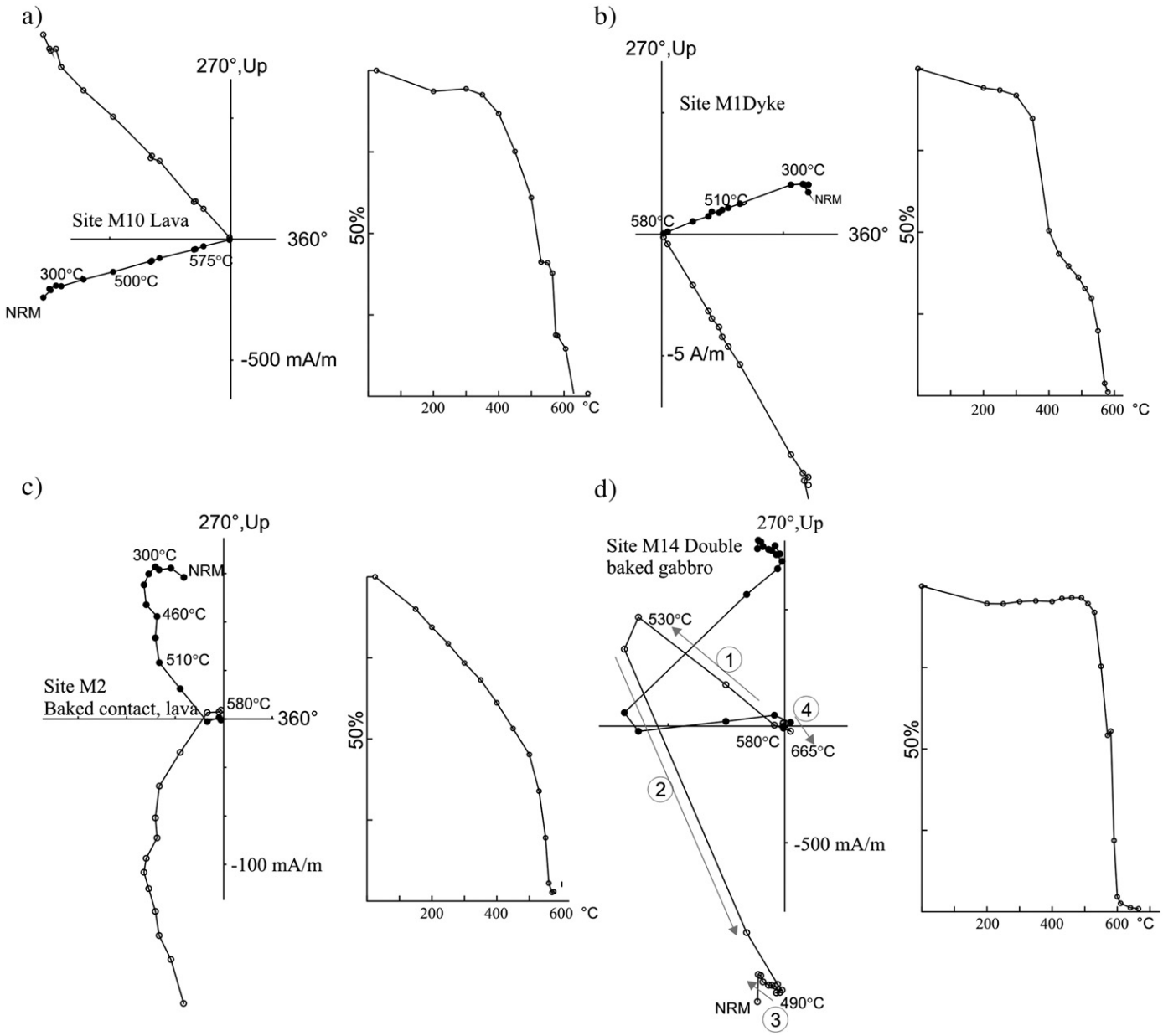


Fig. 2. Orthogonal plots and demagnetisation curves of stepwise demagnetisation data (*in situ* coordinates): (a) typical behaviour of the Mull Plateau Lava Fm., (b) dyke sample from the contact test at site M1, (c) baked lava sample near the dyke at site M2. The curve on the orthogonal plot implies removal of a normal polarity component of NRM related to the dyke followed by removal of a primary reversed polarity component, (d) doubly baked gabbro. Solid (open) symbols represent data projected into the horizontal (vertical) plane.

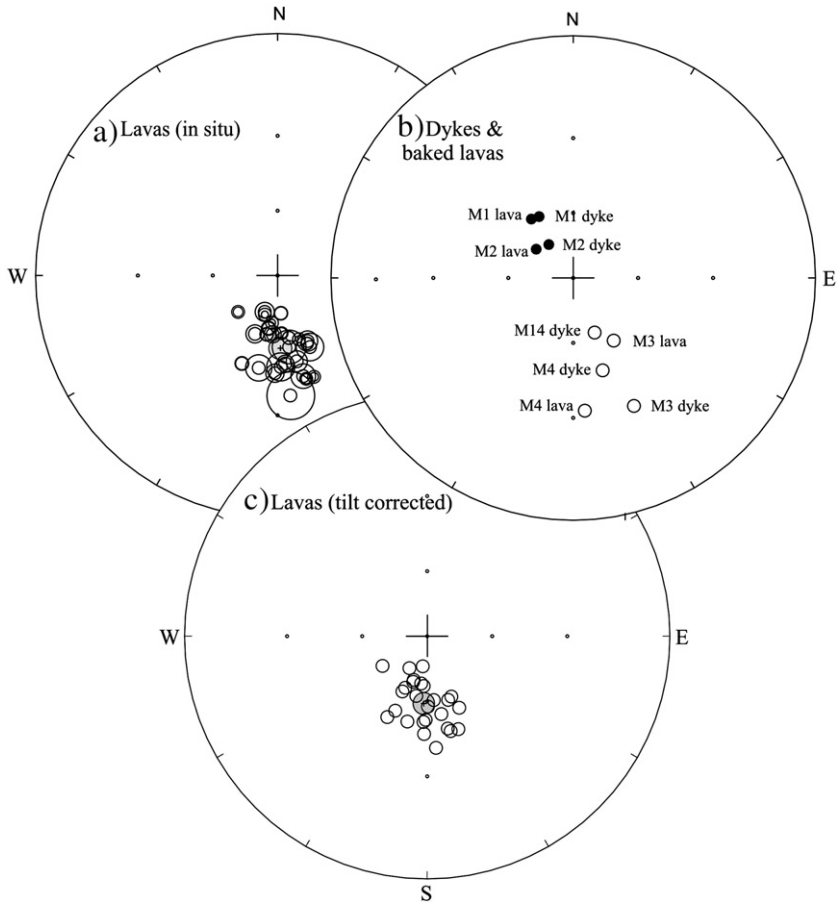


Fig. 3. Equal area projection of flow mean directions for the Mull Plateau Lava Fm. and dykes cutting it (see Table 1). Open (closed) symbols are projected onto the upper (lower) hemisphere. α_{95} confidence limits are shown. (a) *In situ* directions for 26 flows in the MPLF. The overall mean direction for the 26 flows is shaded. (b) Mean directions for dykes and the lava flows they cut (outside baked contacts), (c) flow mean directions corrected for tilt. The overall mean direction is shaded.

overall in-situ site mean direction with declination 177.5° and inclination −57.3° (k=35.9, α_{95}=4.8, N=26; Table 1).

The lava flows in Mull occupy a large-scale gentle syncline (e.g. Bailey et al., 1924). Dips are shallow, usually less than 5°, and in places modified by local faulting. The original dips of the lava flows are unknown but presumably small and less than the dips associated with development of the syncline and fault blocks. Therefore we adjust the in situ remanence directions for folding/faulting by rotating them by amounts that translate the lava flows to the horizontal (Fig. 3c, Table 1). The tilt correction yields a site mean direction of declination 182.9° and inclination −58.9° (k=36.9, α_{95}=4.7, N=26; Table 1). The effect on the mean of site mean direction is a translation of 3.3° of arc and an increase in Fisher's (1953) precision parameter k from 35.9 to 36.9. The tectonic adjustment is minor and the fold test is statistically inconclusive. Nevertheless, since there is clear evidence that the remanence is primary, this small tectonic adjustment to the direction is appropriate. The tilt corrected flow mean directions in Fig. 3c are well clustered. The goodness of fit to the Fisherian distribution for the Mull remanence directions is lower than the critical values (M_u=1.038<1.207, M_e=0.777<1.094) and the null hypothesis that the distribution is Fisherian cannot be rejected at the 95% confidence level (Fisher et al., 1987) and there is no indication of field excursions or transitional fields in the data set.

The degree of paleosecular variation (PSV) is usually expressed in terms of the angular standard deviation (ASD) of virtual geomagnetic poles (VGP) with the implicit assumption that the distribution is Fisherian (Fisher, 1953). The goodness of fit to the Fisherian distribution for the Mull VGP's is lower than the critical values (M_u=0.91<1.207, M_e=0.672<1.094) and the null hypothesis that the distribution of VGP's is Fisherian cannot be rejected at the 95% confidence level (Fisher et al., 1987). The ASD of VGP's, corrected for within flow dispersion Sw=4.3° (McFadden et al., 1991) is 17.5 with a 95% confidence limit of 14.5≤ASD≤21.2 (Cox, 1969). This ASD is in remarkable agreement the PSV prediction of the G-model of McFadden et al. (1991) which indicates an ASD value of ~17 for this time and paleolatitude. We therefore conclude that determination of the overall mean direction from tilt-corrected flow results satisfactorily averages out paleosecular variation.

Using Ben More as a geographic centre (56.25°N, 6.01°E), we calculated a paleomagnetic pole position for the MPLF corresponding to this overall mean direction at age 59±0.2 Ma (weighted mean age of the two dated levels of Chambers and Pringle (2001)). The pole coordinates are 73.3°N, 166.2°E (dp=5.2, dm=7.0; star in Fig. 4, Table 2).

6. Paleomagnetic poles from the NAIP

We have compiled and evaluated paleomagnetic results reported in the peer reviewed literature from the NAIP and consider them in two geographically separated groups: (1) British Tertiary Igneous Province and Faeroe Islands (Eurasian plate), (2) East Greenland and West Greenland (Greenland plate). Our compilation of poles is listed in Table 2 with our new Mull results and illustrated in Fig. 4. As a general rule we favoured the results of newer studies over old, studies with thorough

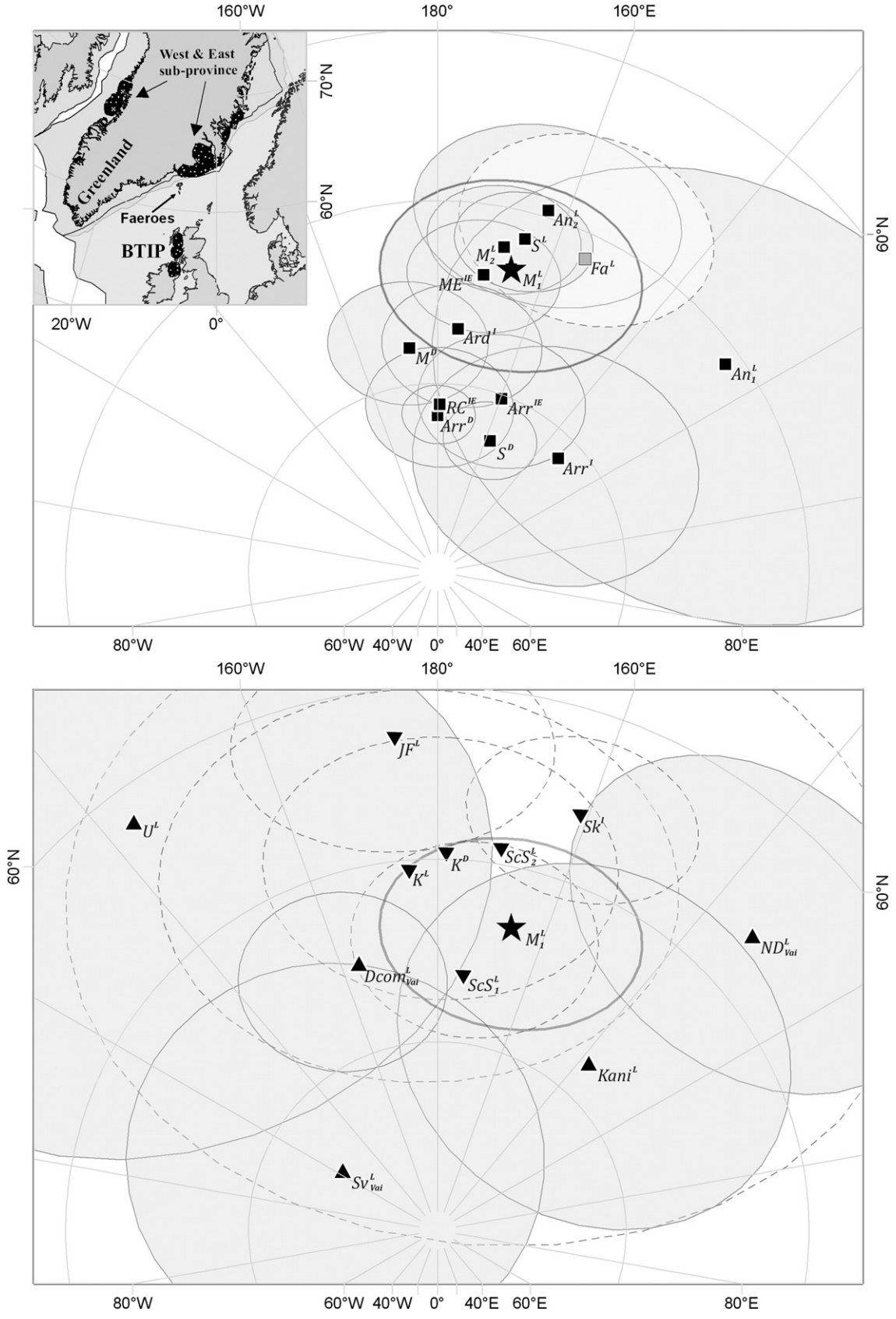


Fig. 4. Selected paleomagnetic poles for the North Atlantic Igneous Province (see Table 2). Greenland (inset) and its data (bottom) are rotated 12.2° counter clockwise around an Euler pole at 48.3°N, 124.5°N to restore them to their pre-breakup positions with respect to Europe (Gaina et al., 2002). The overlap (predrift extension) is shown in transparent (inset) Jan Mayen and Rockall are not shown. Continent-ocean boundaries are taken from Gaina et al. (2007). Poles from each sub-province (top and bottom) are symbolized as follows: BTIP = black squares and Faeroe Islands = grey square with dashed 95% confidence ellipse (top). West Greenland = upright triangles; East Greenland = inverted triangles with dashed 95% confidence ellipsis (bottom); Poles labelled after Table 2. The new Mull pole is symbolized as a star (top and bottom).

Table 2
Selected paleomagnetic poles for the NAIP

Sub province	Rock unit	Label	Age (Ma)	sLat (°N)	sLong (°E)	N/n	D	I	k	α_{95}	pLat (°N)	pLong (°E)	60Ma (°N)	60Ma (°E)	dp	dm
British Isles	Thisstudy:Mull,lavas(1,100%,N/A,N/A,R)	M_1^L	60.5 – 58	56.25	−6.1	26/319	182.9	−58.9	36.9	4.7	73.3	166.2	–	–	5.2	7.0
	Mull, lavas (2, 100%, 1659, 2, S)	M_2^L	60.5–58	56.25	−6.1	78/492	182.1	−57.5	32.0	2.8	71.9	167.2	–	–	3.0	4.1
	Mull, dykes (3, 75%, 2494, 3, R)	M^D	60 – 58	56.6	−6.2	154/530	176.0	−63.0	17	2.7	78.0	187.0	–	–	3.0	4.0
	Antrim, lavas (4, 100%, 2496, 3, R)	An_1^L	61–58	55.1	−6.4	19/79	198.5	−59.7	11.5	10.3	71.0	125.8	–	–	12.1	13.6
	Antrim, lavas (5, 100%, 3026, 2, S)	An_2^L	61–58	55.05	−6.05	25/–	184.7	−54.3	33.3	5.1	69.6	162.9	–	–	5.0	7.0
	Skye, lavas (6, 100%, 2506, 2, S)	S^L	59–58	57.4	−6.3	90/344	183.4	−58.4	35.8	2.5	71.5	165.2	–	–	2.8	3.8
	Skye, dykes (7, 80%, 75, 2, S)	S^D	59–55	57.1	−5.9	409/1636	183.2	−67.1	–	1.5	82.5	158.0	–	–	2.1	2.5
	Arran, dykes (8, 82%, 200, 2, S)	Arr^D	61–58	55.6	−5.2	413/–	179	−65.2	36.8	1.2	81.7	179.8	–	–	1.6	1.9
	Arran, intrusives (9, 40%, 6090, 2, S)	Arr^I	61–58	55.5	−5.2	10/100	8.8	66.2	104.6	4.7	81.2	133.2	–	–	6.3	7.7
	Arran, intrusives and extrusives (10, 75%, 8718, 2, S)	Arr^{IE}	61–58	55.5	−5.2	165/–	183.7	−64.2	–	2.82	80.2	159.6	–	–	3.6	4.5
	Muck and Eigg igneous rocks (11, 97%, 145, 3, R)	ME^{IE}	63–53	56.9	−6.2	133/524	181.0	−59.9	21.5	2.7	73.9	171.1	–	–	3.1	4.1
	Rhum and Canna, intrusives and extrusives (12, 95%, 68, 2, S)	RC^{IE}	60–58	57.0	−6.5	107/445	178.7	−65.9	33.5	2.4	81.1	179.1	–	–	3.2	3.9
	Ardnamurchan, intrusives (13, 100%, 146, 3, R)	Ard^I	61–58	56.7	−6.2	62/484	180.0	−63.0	–	2.7	77.0	175.0	–		3.3	4.2
Faeroes	Faeroes, lavas (14, 70%, 8901, 3, R)	Fa^L	57–55	61.9	−6.9	43/456	187.7	−60.9	24.5	4.5	71.4	154.7	–	–	6.0	6.0
East Greenland	Scoresby Sund lavas (15, 100%, N/A, 2, S)	ScS_1^L	56–54	70.08	−25.59	5/125	166.5	−66.3	368.9	4.0	69.7	181.4	76.7	174.7	5.8	5.8
	Scoresby Sund lavas (16, 100%, 3404, 2, S)	ScS_2^L	56–54	70.0	−25.0	28/157	168.0	−62.0	–	15.0	63.0	174.0	69.1	170.9	16.0	22.0
	Jackobsen Fjord lavas (17, 100%, 1088, 2, S)	JF^L	60–59	68.2	−31.0	39/200	157.0	−56.0	–	6.0	55.8	177.0	62.8	185.2	6.2	8.6
	Kangerdlugssuaq lavas (18, 100%, 1584, 2, S)	K^L	58–56	68.05	−32.0	4/22	158.5	−62.8	105	8.9	63.4	185.1	70.5	184.8	11	15
	Skaergaard intrusion (19, 100%, 612, 2, S)	Sk^I	56–55	68.2	−31.7	3/30	170.0	−59.0	40.0	4.2	61.0	165.0	66.1	161.4	4.6	6.2
	Kangerdlugssuaq dykes (17, 100%, 1088, 2, S)	K^D	55–54	68.2	−31.0	11/66	160.9	−62.0	57.5	6.1	62.9	180.4	69.6	179.0	7.3	9.5
West Greenland	Nuussuaq lavas (Kanisut Mb.) (20, 100%, N/A, 3, R)	$Kani^L$	55–53	70.71	−54.55	20/134	164.0	−70.7	27.8	6.3	74.6	159.4	78.0	137.8	10.3	10.3
	Nuussuaq and Disko lavas (Vaigat Fm.) (21, 25%, N/A, 3, S)	ND_{Vai}^L	61–60	70.34	−54.87	14/99	349.8	62.6	34.1	6.9	64.8	141.5	66.6	133.0	9.2	9.2
	Svartenhuk lavas (Vaigat Fm.) (20, 100%, N/A, 3, S)	Sv_{Vai}^L	61–60	71.6	−54.1	10/167	139.9	−77.7	65.6	6.0	76.2	217.9	84.1	237.7	9.8	9.8
	Disko lavas combined (Vaigat Fm.) (22, 95%, 2614 and 2782, 2, S)	$Dcom_{Vai}^L$	61–60	70.0	−53.5	81/–	140.7	−71.1	–	3.2	67.5	195.0	75.2	196.3	4.9	5.6
	Ubekendt Ejland lavas (23, 100%, 2854, 2, S)	U^L	52–50	71.1	−54.0	6/32	121.3	−66.3	36.4	11.2	54.0	208.0	62.1	216.6	15.0	18.0

Rock unit: in parentheses, article reference number in list below, % reversed, global database result ID, database demagnetization code, R = documented result, S = supporting data. Label: labels used in Figs. 4 to 6. Ages are from the following sources: Mull — Mussett (1986), Chambers and Pringle (2001); Antrim—Thompson (1985); Skye — Bell and Williamson (2002); Arran — Chambers and Fitton (2000); Muck, Eig, Rhum and Canna — Chambers et al. (2005); Faeroe Islands — Storey et al. (2007); West Greenland — Storey et al. (1998), Sinton and Duncan (1998); East Greenland — Hansen et al. (2002). sLat/sLong: study latitude/longitude; *N*/*n*: number of sites/samples; *D*/*I*: mean remanence declination/inclination; *k*: Fisher's (1953) precision parameter; α_{95}: half angle of cone of 95% confidence; pLat/pLong: pole latitude/longitude; 60 Ma (°N and °E): Greenland poles rotated into pre-Atlantic positions with respect to Europe; dp/dm: semi axes of the ovals of 95% confidence in pole positions. References for paleomagnetic data: (1) this study, (2) Hall et al. (1977), (3) Ade-Hall et al. (1972), (4) Løvlie et al. (1972), (5) Wilson (1970), (6) Wilson et al. (1972), (7) Wilson et al. (1982), (8) Dagley et al. (1978), (9) Mussett et al. (1987), (10) Hodgson et al. (1990), (11) Dagley and Mussett (1986), (12) Dagley and Mussett (1981), (13) Dagley et al. (1984), (14) Riisager et al. (2002), (15) Tarling et al. (1988), (16) Tarling (1967), (17) Faller and Soper (1979), (18) Faller (1975), (19) Schwarz et al. (1979), (20) Riisager et al. (2003b), (21) Riisager et al. (2003a), (22) Athavale and Sharma (1975) and (23) Tarling and Otulana (1972).

laboratory analysis, based on many sampling sites, carried out on multiple lava flows where paleohorizontal can be estimated, and on multiple intrusions where some degree of time averaging can be expected. We might have been generous including some poles and harsh excluding others, but the pole position groupings and trends in the different geographic regions are nonetheless clear. Pole labels indicate place and rock type (D = dykes, L = lavas and IE = intrusives and extrusives). Due to the paucity of modern data we include data from a number of older studies based on limited laboratory analysis. These are collectively termed as "supporting data".

6.1. Eurasian plate

Similar pole positions have emanated from the major lava sequences of Antrim ($An_{1\ and\ 2}^L$), the Isle of Mull (M_2^L), the Isle of Skye (S^L) and of the Faeroe Islands (Fa^L Fig. 4, Table 2). Results from intrusive bodies of similar age resemble these (Arr^D, Arr^I, Arr^{IE}, S^D, Ard^I, M^D, RC^{IE}, ME^{IE}). The newer pole from the Faeroes lava sequence of Riisager et al. (2002) supersedes poles from the older studies by Tarling (1970) and Løvlie and Kvingedahl (1975).

The position of our new pole (labelled M_1^L in Fig. 4) is indistinguishable from the bulk of BTIP paleomagnetic data and entirely consistent with that of Hall et al. (1977) on the same lava sequence and a study of Ade-Hall et al. (1972) on the dyke swarms of Mull (Table 2; poles M_2^L and M^D in Fig. 4).

6.2. Greenland plate

The paleomagnetic poles derived from Greenland in our list predate the opening of the North Atlantic Ocean. Therefore to compare them with poles from the Eurasian plate, we rotated the pole positions from Greenland about an Euler pole that restores Greenland to its 60 Ma position relative to Eurasia (Table 2, Fig. 4). The rotation used is 12.2° counter-clockwise about an Euler pole at 48.3°N, 124.5°E, interpolated between rotations for Chron 25 (55.9 Ma) and 31 (67.7 Ma) of Gaina et al. (2002) obtained by statistically fitting ocean floor magnetic anomalies in the North Atlantic and Baffin Bay.

6.2.1. East Greenland

Six poles were selected from the available paleomagnetic data for East Greenland: two from lavas in Scoresby Sund ($ScS_{1\ and\ 2}^L$); two from lavas in the Kangerdlugssuaq area (K^L and K^D); one from lavas near Jacobsen Fjord (JF^L) and one from the Skaergaard intrusion (Sk^I; Fig. 4, Table 2). The majority of these studies are old and all employed limited demagnetisation methods in the identification of characteristic remanence. Nevertheless, as shown in Fig. 4, poles from this region are in general agreement.

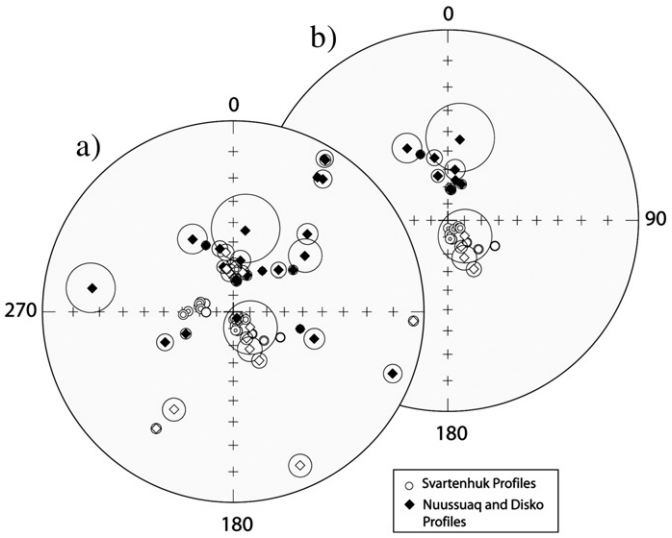


Fig. 5. Remanence directions for the Vaigat Fm. (Riisager et al., 2003a,b). (a) Entire and (b) final dataset with outliers removed. The Svartenhuk profiles are symbolized as circles and squares are used for the Disko/Nuussuaq profiles.

6.2.2. West Greenland

Paleomagnetic results from West Greenland (Table 2, Fig. 4) include those from recent studies by Riisager et al. (2003b) ($Kani^{L}$, Sv^{L}_{Vai}) and Riisager et al. (2003a) (ND^{L}_{Vai}). Results from older studies include a combined pole for the Vaigat Fm. in Kristjansson and Deutsch (1973) and Athavale and Sharma (1975) ($Dcom^{L}_{Vai}$), and a pole from Tarling and Otulana (1972) for the younger Ubekendt Ejiland lavas (U^{L}). Fig. 4 shows that the overall assemblage of pole positions for West Greenland is widely scattered compared with data from the other regions.

Detailed studies in the Vaigat Fm. at Svartenhuk and Disko/Nuussuaq (Riisager et al., 2003a,b) revealed widely varying directional trends (Fig. 5), attributed by the authors to the presence of transitional and/or excursional field directions. We found it difficult to extract excursional or transitional fields from the complex pattern of directional groups (Fig. 5). Nevertheless, having removed directional trends associated with transitional fields, Riisager et al. (2003b)'s mean pole positions for the Vaigat Formation at Svartenhuk and Disko/Nuussuaq, remain clearly distinguishable at the 95% confidence level (76.2°N, 217.9°E, A_{95}=9.8 for Svartenhuk and 64.8°N, 141.5°E, A_{95}=9.2 for Disko/Nuussuaq, Fig. 4). Despite this, Riisager et al. (2003b) calculated an overall mean pole for the Vaigat Fm. that then resembles the result from the younger Kanisut Member and, when adjusted for opening of the Atlantic, the pole from the Faeroe Islands lava sequence (Riisager et al., 2002). Finding no reason to combine the strictly different mean directions from Svartenhuk and Disko/Nuussuaq we adopt here a conservative approach and add the two separate poles for the Vaigat Fm. to Table 2 (Sv^{L}_{Vai}=Svartenhuk, ND^{L}_{Vai}=Disko/Nuussuaq) and Fig. 4.

7. Comparison of poles from the NAIP

Fig. 4 shows poles from the different parts of the NAIP. Results from the BTIP are shown by squares and a star indicates the new Mull result. The Mull result presented here confirms previously published paleopole positions and we conclude that results from the BTIP can be used with confidence. This is contrary to the conclusion of Riisager et al. (2002) and Riisager et al. (2003b).

To simplify the comparison of results for the different parts of NAIP, we generate mean poles for the BTIP, East Greenland and West Greenland and display them with the Faeroe Islands pole in Fig. 6 in European co-ordinates (also see Table 3). Fig. 6 also shows the overall mean pole for all data from the NAIP (75.0°N, 167.8°E, k=84.3, A_{95}=3.2, N=25, Table 3).

The figure clearly shows general agreement between mean poles from the different sub-provinces of the NAIP, indicating overall consistency in the paleomagnetic record throughout the LIP. We do not assign any geological significance to differences between the mean poles from the sub-provinces at this stage given the variable reliability of the paleomagnetic data underlying the mean poles and given uncertainties in the ages of some of the rock units. This observation is contrary to that of Riisager et al. (2002) and Riisager et al. (2003b) who proposed that data from the BTIP differed from the rest and concluded that the BTIP data were unreliable. We can now confirm the reliability of key data from the BTIP and reject the concept of large-scale paleomagnetic inconsistency across the NAIP.

Our new overall mean pole for the NAIP (Table 2) is compared with the apparent polar wander (APW) path for Eurasia of Torsvik et al. (in review) in Fig. 6. To facilitate this comparison we first extracted all NAIP data from Torsvik et al's data compilation and generated a revised APW path using running mean interpolation with a window size of 10 Ma and window interval of 10 Ma. The NAIP pole lays some 6.1°of arc from its time equivalent position in the APW path (60 Ma). Given uncertainties involved in determination of the NAIP pole, the age range of the results incorporated into this pole, and uncertainties and inevitable smoothing in the APW path, we cannot demonstrate a significant difference between the pole and its expected position in the path. Therefore we are confident that the NAIP pole (Table 3) is a sound estimate of the generalised location of the paleomagnetic pole with respect to Eurasia during eruption of the NAIP (~62 Ma–52 Ma).

8. Discussion

We are aware of a number of investigations focusing on the detailed behaviour of the geomagnetic field at the time of eruption of the NAIP, including a recent contribution by Moreau et al. (2007). Moreau et al. (2007) stacked Paleocene–Eocene magnetostratigraphies from around the world and identified an APW loop that they attributed to true polar wander (TWP). This is only possible through detailed examination of precisely dated magneto-stratigraphic records. The question arises as to whether such a loop is evident in our selection of paleomagnetic data from the NAIP. A rapid examination of the data sets listed in Table 2 is enough to show that the available paleomagnetic data for the NAIP are either not detailed enough, or well enough dated, to shed light on smaller scale APW behaviour like that forwarded by Moreau et al. (2007).

One clear feature of Fig. 6, however, is worth further discussion at this stage. It is quite noticeable that the West Greenland data are more dispersed than data from other provinces in the NAIP. There can be a number of reasons for this but our attention is particularly attracted to a directional difference between modern paleomagnetic data from spatially separated but contemporaneous lava sequences reported by Riisager et al. (2003b). We have already observed that their mean pole for the Vaigat Fm. at Svartenhuk differs from the pole for the same formation at Disko/Nuussuaq — after transitional fields and excursions have been removed (Fig. 5, Table 2). We will now address the question of whether relative tectonic rotation within West Greenland can explain differences in paleomagnetic directions.

The Labrador Sea and Baffin Bay spreading systems are offset by the Ungava Fault Zone (UFZ), which is essentially is a left lateral transform (Funck et al., 2007). The UFZ, or its splays, most likely extend into the Disko–Svartenhuk area (Chalmers et al., 1999) and connect with the Itilli fault, which cuts the lava sequences at Hareøen, Nussuuaq and most probably Ubekendt Ejland. The area around Disko–Svartenhuk is highly dissected by faults and was subjected to protracted periods of rifting between the Late Cretaceous and Eocene

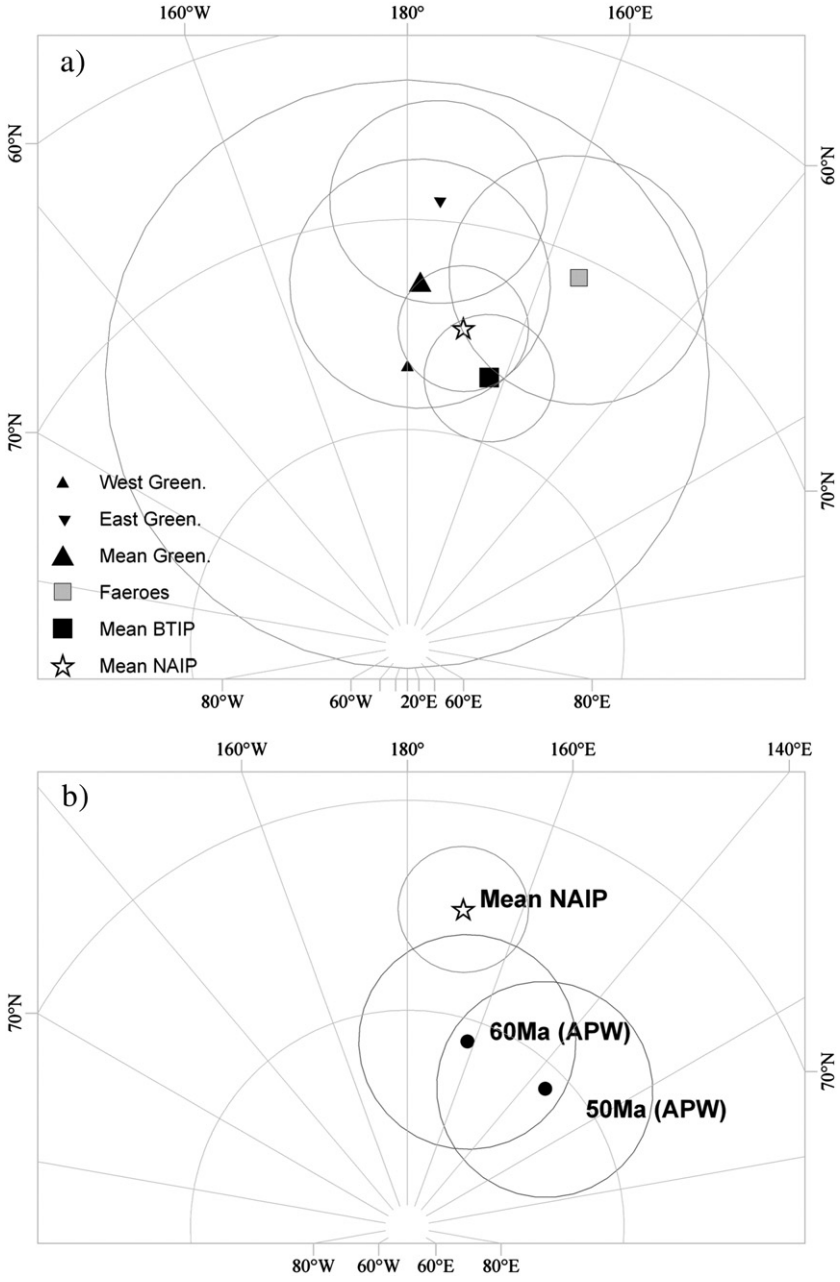


Fig. 6. (a) Mean poles for each of the NAIP sub-provinces. (b) Overall mean pole together with 60 Ma and 50 Ma pole positions extracted from the apparent polar wander for Eurasia (modified after Torsvik et al., in review).

(Chalmers et al., 1999, Larsen and Pulvertaft, 2000, Geoffroy et al., 2001).

The lava sequences themselves have shallow dip and these dips were corrected paleomagnetic directions be related to near vertical axis rotation brought about by movements on the aforementioned fault systems, or related but poorly exposed fault systems? We notice that lineaments on Ubekendt Ejland differ in trend by 14° clockwise from similar lineaments at Svartenhuk. Could this be related to sinistral motion across a fault system in the region, matching the sinistral sense of displacement across the UFZ?

A vertical axis rotation of 15–25° of Disko/Nuussuaq (counter-clockwise) and Svartenhuk (clockwise) is required to translate paleomagnetic poles from the two regions to their closest positions with respect to the East Greenland mean pole (Table 3). The same rotations are required to translate the two poles into coincidence with the slightly younger pole for the overlying syn-break-up Kanisut Formation. Without stronger structural geological evidence for vertical axis rotation within the Disko–Svartenhuk region, rotations as large as those required to reconcile the paleomagnetic data cannot be substantiated.

Table 3
Mean poles for the NAIP

Mean	Label	Lat (°N)	Long (°E)	*N*	*k*	A_{95}
West Greenland	West Green.	76.9	180.4	5	30.1	14.2
East Greenland	East Green.	69.4	176.1	6	197.4	4.8
Greenland	Mean Green.	72.8	177.6	11	54.9	6.2
Faeroes	Faeroes	71.4	154.7	1	24.5	6.0
BTIP	Mean BTIP	76.9	163.9	13	172.5	3.2
NAIP	Mean NAIP	75.0	169.9	25	84.3	3.2

9. Conclusions

New paleomagnetic results from the Mull Plateau Lava formation have lead to a new paleomagnetic pole for Eurasia for the weighted age of 59 ± 0.2 Ma with coordinates 73.3°N, 166.2°E (dp = 5.2, dm = 7.0). This pole conforms and refines earlier paleomagnetic results from the British Tertiary Igneous Province.

A thorough re-examination of a compilation of previously published data allows comparison of this new pole with the rest of NAIP. We found an internal consistency between the data of the Eurasian part of NAIP (BTIP and Faeroe), which are in general agreement with the East Greenland data (rotated to Eurasia). However, new and old data from the West Greenland do not show an internal consistency and are associated with high dispersion especially for the pre-break up lava of the Vaigat Fm. We speculate on the origin of this complex pattern and raise the question of undetected more complex tectonic history linked with movements along the Ungava Fault complex.

All data combined, give a pole for the North Atlantic Igneous Province with coordinates 75.0°N, 169.9°E (N = 25, K = 84.3, A_{95} = 3.2). Removing data from West Greenland have minor impact on the overall pole position (74.5°N, 167.8°E) only its estimators (N = 20, K = 133.3, A_{95} = 2.8).

Acknowledgements

We thank Statoil for financial support during the fieldwork of summer 2004 and early stages of the investigation. M.G. acknowledges the Norwegian Research Council for funding the rest of this research. Donald Ramsey provided invaluable assistance in the field and we thank him for that. Also, our thanks go to Harald Walderhaug and Reidar Løvlie for granting us access to the facilities in the paleomagnetic laboratory in Bergen, and to Henry Emeleus for valuable input on the geology of Mull. Randy Enkin kindly provided a very helpful review that improved the paper immensely. We also greatly appreciate laboratory assistance from Tom Jacobsen, Bengt Johansen and Britt Vongraven of the Geologic Survey of Norway.

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